PERFECT (ADVERSARIAL) INDISTINGUISHABILITY:

We imagine a "challenge" game between a passive adversary A and the encryption scheme Π:

- A picks two messages $m_0, m_1 \in \mathcal{M}$
- The challenger:
  - runs $k \leftarrow Gen()$
  - flips a fair coin $b \in \{0,1\}$
  - computes the challenge ciphertext
    $c \leftarrow Enc_k(m_b)$ and gives c to A.
- A outputs a guess $b' \in \{0,1\}$.
- A wins if $b' = b$

  we write $Pr[Privk^{eav}_{A,\Pi} = 1] = Pr[b' = b]$

Because b is uniform, any A can always succeed with probability atleast $1/2$ by random guessing.

We say Π is perfectly indistinguishable if

$$\forall A : Pr[b' = b] = 1/2$$

If, upon seeing c, A can predict b with probability strictly greater that $1/2$, then c must carry some information that distinguishes m_0 from m_1.

Conversely, if c is distributed exactly the same whether you encrypted m_0 or m_1, then no adversary (no matter how powerful) can do better than blind guessing.

Contrapositive of previous statement

Statement A: If adversary does better than guessing, the c reveals some info. (i.e., the distribution differs)

Statement B: If c does not reveal any distinguishing information (i.e., ciphertext distributions are identical), then the adversary cannot do better than guessing.

Encryption scheme Π is perfectly secret if and only if it is perfectly indistinguishable.

Let Π denote the Vigenère cipher for the message space of two character strings with key period chosen uniformly from $\{1,2\}$.

The adversary chooses:

- $m_0 = aa$
- $m_1 = ab$

Upon receiving ciphertext $c = c_1 c_2$, the adversary does,

if $c_1 = c_2$, output 0 (guess $b=0$)

otherwise, output 1 (guess $b=1$)

Goal is to compute $Pr[PrivK^{eav}_{A,\Pi} = 1]$ and show that it's greater than $1/2$.

& the probability is given by,

$$Pr[PrivK^{eav}_{A,\Pi} = 1] = 1/2 \cdot Pr[A \text{ outputs } 0 \mid b=0] + 1/2 \cdot Pr[A \text{ outputs } 1 \mid b=1]$$

where b is the uniform bit determining which message gets encrypted.

When $b=0$ (so m_0 is encrypted) then $c_1=c_2$ if either

a key period 1 is chosen

So same key letter applied to both letters.

key $k=k_1k_1$

$c_1=a+k_1$, $c_2=a+k_1$

so $c_1=c_2$ always

This happens with probability $1/2$ since perior is choosen uniformly from $\{1,2\}$

So $Pr[c_1=c_2 \mid \text{key period}=1]=1$

and $Pr[\text{Key period}=1]=1/2$

$\Rightarrow 1/2 \cdot 1 = 1/2$

a key period 2 is chosen

key $k=k_1k_2$

$c_1=a+k_1$, $c_2=a+k_2$

$c_1=c_2 \Longleftrightarrow k_1=k_2$

There are 26 choices for k_1 and k_2 respectively.

$\therefore$ $26\times 26 = 676$ possible keys.

But out of which only 26 of them satisfy $k_1=k_2$

So $Pr[k_1=k_2]=\frac{26}{676}=1/26$

$Pr[\text{key period}=2]=1/2$

$\Rightarrow 1/2 \cdot 1/26$

$\therefore$ Total probability is

$$Pr[c_1=c_2 \mid b=0]=\frac{1}{2}+\frac{1}{2}\cdot\frac{1}{26} = \frac{1}{2}(1+1/26)=1/2(27/26)$$

$$\therefore Pr[A \text{ outputs } 0 \mid b=0] = 27/52 \approx 0.52$$

When $b=1$, message m_1 is encrypted.

A outputs 1 if $c_1 \neq c_2$.

Key period $= 1$

i.e., key $k = k_1 k_1$

If we encrypt ab with key kk, is it possible that the expected result is XX?

$C_1 = a + k_1 = 0 + k_1 = k_1 \mod 26$

$C_2 = b + k_1 = 1 + k_1 = 1 + k_1 \mod 26$

Are $c_1 = c_2$?

i.e., $k_1 = k_1 + 1 \mod 26$?

No. so in every case $c_1 \neq c_2$.

So $Pr[C_1 = C_2 \mid \text{key period} = 1] = 0$

Key period $= 2$

i.e., key $k = k_1 k_2$

$c_1 = a + k_1$, $c_2 = b + k_2$

$c_1 = 0 + k_1$, $c_2 = 1 + k_2$

$= k_1$

For $c_1 = c_2$, we need,

$k_1 = k_2 + 1 \mod 26$

$\Rightarrow k_1 - k_2 = 1 \mod 26$

There are 26 possible values for k_2 and for each one, there is exactly one corresponding k_1 satisfying the condition.

So, total key pairs $= 676$

Matching pairs $= 26$ one for each k_2

So, $Pr[k_1 = k_2] = \frac{26}{676} = \frac{1}{26}$

& $Pr[\text{key period} = 1] = Pr[\text{key period} = 2] = 1/2$

$\therefore Pr[c_1 = c_2 \mid \text{key period} = 2] = 1/2 \cdot 1/26$

So $\Pr[A \text{ outputs } 1 \mid b=1] = 1 - \Pr[A \text{ outputs } 0 \mid b=1]$

$$= 1 - \tfrac{1}{2} \cdot \tfrac{1}{26}$$

$$= 1 - \tfrac{1}{52} = \tfrac{51}{52} \approx 0.98$$

$\therefore$ $\Pr[\mathrm{PrivK}^{eav}_{A,\Pi} = 1] = \tfrac{1}{2} \cdot \tfrac{27}{52} + \tfrac{1}{2} \cdot \tfrac{51}{52} = \dfrac{1}{2 \cdot 52}(27 + 51)$

$$= \frac{\cancel{78}^{\,3}}{2 \cdot \cancel{52}_{\,2}} = \tfrac{3}{4} = 0.75 > \tfrac{1}{2}$$

$\Longrightarrow$ This scheme is not perfectly indistinguishable.